



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Omar Hani Ibrahim Bishr
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Data Collection through API
- Data Collection with Web Scraping
- Data Wrangling
- Exploratory Data Analysis with SQL
- Exploratory Data Analysis with Data Visualization
- Interactive Visual Analytics with Folium
- Machine Learning Prediction

•Summary of all results

- Exploratory Data Analysis result
- Interactive analytics in screenshots
- Predictive Analytics result

Introduction

- Project background and context
 - This project aims to predict the landing success of SpaceX Falcon 9's first stage. SpaceX advertises Falcon 9 rocket launches at a cost of \$62 million, significantly lower than competitors, who charge upwards of \$165 million per launch. This cost reduction is largely attributed to SpaceX's ability to reuse the first stage of their rockets. By determining whether the first stage will land successfully, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.
- Problems that need answers
 - Factors that determine a successful land
 - Predicting whether a land will be successful or not

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data was collected using SpaceX API and web scraping from Wikipedia website
- Data wrangling
 - Data was wrangled and cleaned to prepare it for queries ,visualizations and training ML models
 - e.g., one-hot encoding was applied on categorical features to prepare them for teaching ML models and class column was created to represent success or failure in first stage landing
- Exploratory data analysis (EDA) was performed using visualization and SQL
- Interactive visual analytics were made using Folium and Plotly Dash
- Predictive analysis was performed using classification models

Data Collection

- The data was collected from:
 - Using an API and a static response json object for consistency
 - By extracting a HTML table from a Wikipedia page using web scrapping
 - More data sets were provided and created through the process of the project

1- Data Collection

i/p: SpaceX API json object
o/p: dataset_part_1.csv

2- Web Scrapping

i/p: Wikipedia HTML table
o/p: spacex_web_scraped.csv

3- Data Wrangling

i/p: dataset_part_1.csv
o/p: dataset_part_2.csv

4- EDA SQL

i/p: Spacex.csv

5- EDA Data Visualization

i/p: dataset_part_2.csv
o/p: dataset_part_3.csv

6- Folium Maps

i/p: spacex_launch_geo.csv

6- Plotly Dashboard

i/p: spacex_launch_dash.csv

7- Machine Learning

i/p: dataset_part_2.csv
i/p: dataset_part_3.csv

Data Collection – SpaceX API

- Data was available through SpaceX API
- A static copy of the response was provided to make the results consistent
- Data was extracted from the response json object
- Then it was turned into a pandas dataframe for further analysis
- **Notebook link:** [Applied-Data-Science-Capstone/jupyter-labs-spacex-data-collection-api.ipynb](https://github.com/OmarHani4306/Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb) at main · OmarHani4306/Applied-Data-Science-Capstone

1- Send Get request to the static response API

2- convert the response to a json object then turn it into a pandas dataframe

3- call helper functions on specific columns to generate data

4- construct a dataframe from the generated data

5- Filter the dataframe to only include `Falcon 9` launches

6- fill the missing values in PayloadMass Column with the mean value

7- export the dataframe to a csv file 'dataset_part_1.csv'

Data Collection - Scraping

- Falcon 9 launch records were scraped using BeautifulSoup by extracting the html table from Wikipedia
- Table was parsed and converted into a pandas dataframe
- **Notebook link:** [Applied-Data-Science-Capstone/jupyter-labs-webscraping.ipynb at main · OmarHani4306/Applied-Data-Science-Capstone](#)

1- send an API get request to a snapshot of the table

2- create a BeautifulSoup object from the response

3- collect all relevant column names from the HTML table header

3- collect all relevant column names from the HTML table header

4- Create a data frame by parsing the launch HTML tables

5- export the dataframe to a csv file 'spacex_web_scraped.csv'

Data Wrangling

- Data was loaded from a previously created CSV file
- Basic data analysis was performed identifying column types and percentages of missing values in each attribute
- the number of launches on each site, the number and occurrence of each orbit, the number and occurrence of mission outcome of the orbits were calculated
- a landing outcome label from Outcome column was created and added to the dataframe
- **Notebook Link:** [Applied-Data-Science-Capstone/labs-jupyter-spacex-Data wrangling.ipynb at main · OmarHani4306/Applied-Data-Science-Capstone](#)

1- load data from 'dataset_part_1.csv' to dataframe

2- Identify and calculate the percentage of the missing values in each attribute

3- Identify which columns are numerical and categorical

4- Calculate the number of launches on each site

5- Calculate the number and occurrence of each orbit

6- Calculate the number and occurrence of mission outcome of the orbits

7- Create a landing outcome label from Outcome column

8- Create a landing outcome label from Outcome column and export the dataframe to csv file "dataset_part_2.csv"

EDA with Data Visualization

- Summary of used charts :
 - catplot(FlightNumber vs PayloadMass)
 - catplot(FlightNumber vs LaunchSite)
 - scatterplot(PayloadMass vs LaunchSite)
 - barplot(Orbit vs Class)
 - scatterplot(FlightNumber vs Orbit)
 - scatterplot(PayloadMass vs Orbit)
 - lineplot(year vs Class)
- These charts were used to visualize the relationships between different variables, and obtain insights about how each important variable would affect the success rate
- Notebook Link: [Applied-Data-Science-Capstone/jupyter-labs-eda-dataviz.ipynb](#) at main · OmarHani4306/Applied-Data-Science-Capstone

EDA with SQL

- Summary of performed SQL queries
 - Displayed the names of the unique launch sites in the space mission
 - Displayed 5 records where launch sites begin with the string 'CCA'
 - Displayed the total payload mass carried by boosters launched by NASA (CRS)
 - Displayed average payload mass carried by booster version F9 v1.1
 - Listed the date when the first successful landing outcome in ground pad was achieved
 - Listed the names of the booster which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - Listed the total number of successful and failure mission outcomes
 - Listed the names of the booster versions which have carried the maximum payload mass
 - Listed the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015
 - Ranked the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- [Notebook Link: Applied-Data-Science-Capstone/jupyter-labs-eda-sql-coursera_sqlite.ipynb at main · OmarHani4306/Applied-Data-Science-Capstone](#)

Build an Interactive Map with Folium

- Summary of map objects created and why they were added:
 - Markers: a marker was created for each launch site, and to mark the success/failed launches for each site on the map
 - Circles: a circle was created for each launch site
 - MousePosition : to get coordinates for a mouse over a point on the map
 - Lines: were used to calculate the distance between a launch site to its proximities
- **Notebook Link:** [Applied-Data-Science-Capstone/lab_jupyter_launch_site_location.ipynb](https://colab.research.google.com/github/OmarHani4306/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb) at main · OmarHani4306/Applied-Data-Science-Capstone

Build a Dashboard with Plotly Dash

- Summary of plots/graphs and interactions and why they were added:
 - Pie chart: when all sites are chosen it displays the percentage of success launches by each site, and when one site is chosen it displays the percentage of success and failure for the site
 - Scatter plot: Displays the correlation between payload mass and mission outcome by booster version category for either all sites or for each individual site
 - An input slider is used to set the range for payload mass in the scatter plot
- [Notebook Link: Applied-Data-Science-Capstone/Build a Dashboard Application with Plotly Dash.ipynb at main · OmarHani4306/Applied-Data-Science-Capstone](#)

Predictive Analysis (Classification)

- Models we built using modules such as (train_test_split, LogisticRegression, SVC, Decision TreeClassifier, KNeighborsClassifier) imported from sklearn library
- Models were evaluated based on accuracy
- Models were improved using GridSearchCV to tune hyperparameters and find the best values for them
- Best performing model was determined based on multiple metrics from sklearn library such as (accuracy_score, fbeta_score, jaccard_score, log_loss)
- **Notebook Link** : [Applied-Data-Science-Capstone/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb at main · OmarHani4306/Applied-Data-Science-Capstone](#)

1- Y series was created from class column from dataframe of dataset_part_2.csv

2- X dataframe was created from dataset_part_3.csv

3- X,Y Data was split into training and test sets

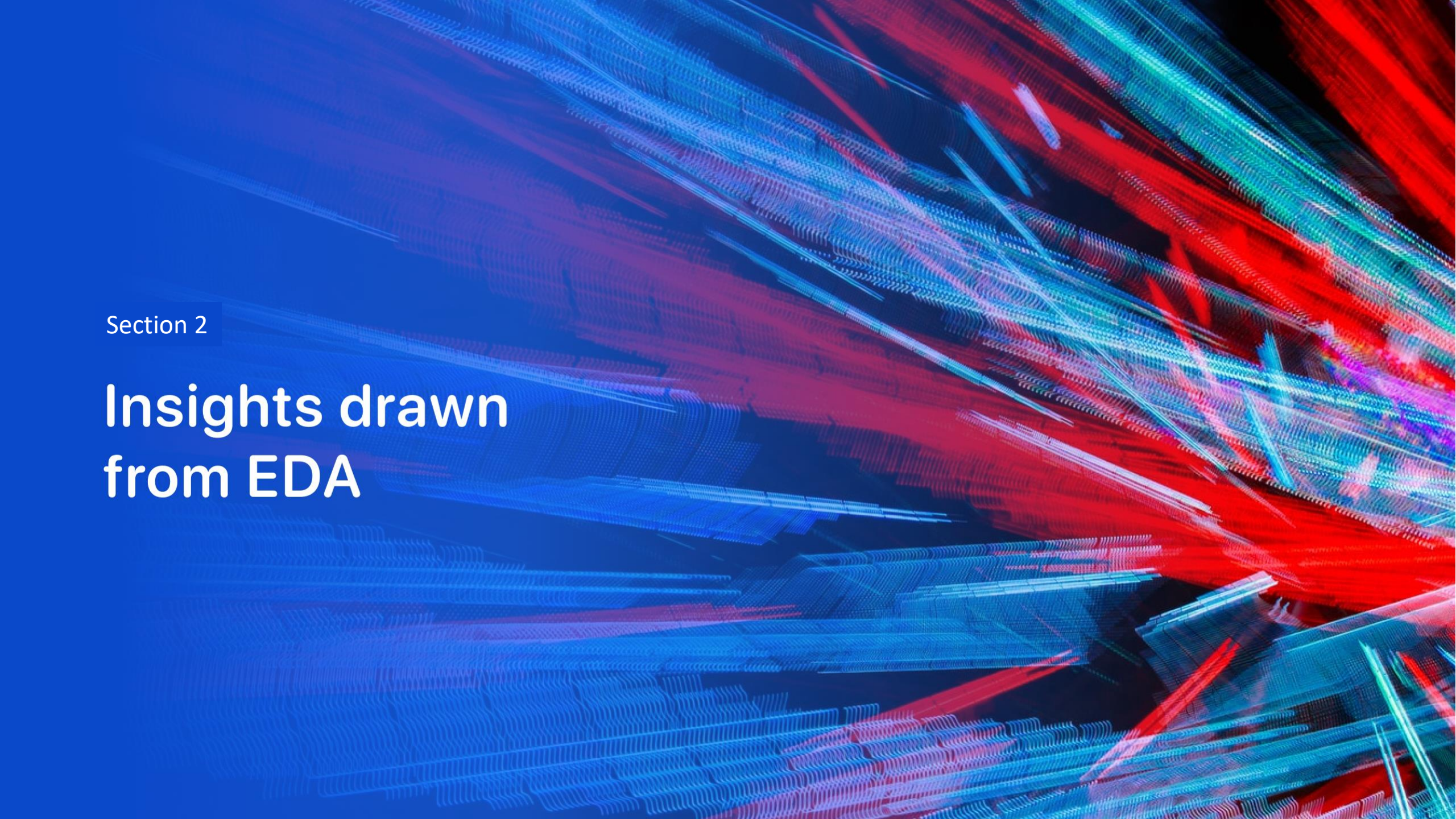
4- Each of the 4 models were trained on the training set, hyperparameters were tuned using gridSearch and evaluated on the test set

5- Each of the 4 models were evaluated based on their accuracy performing on test sets

6- best model was determined based on different multiple metrics

Results

- Exploratory data analysis results
 - Plots and SQL Queries
- Interactive analytics results
 - Interactive folium maps
 - Interactive plotly dash dashboard
- Predictive analysis results
 - Predictive ML models

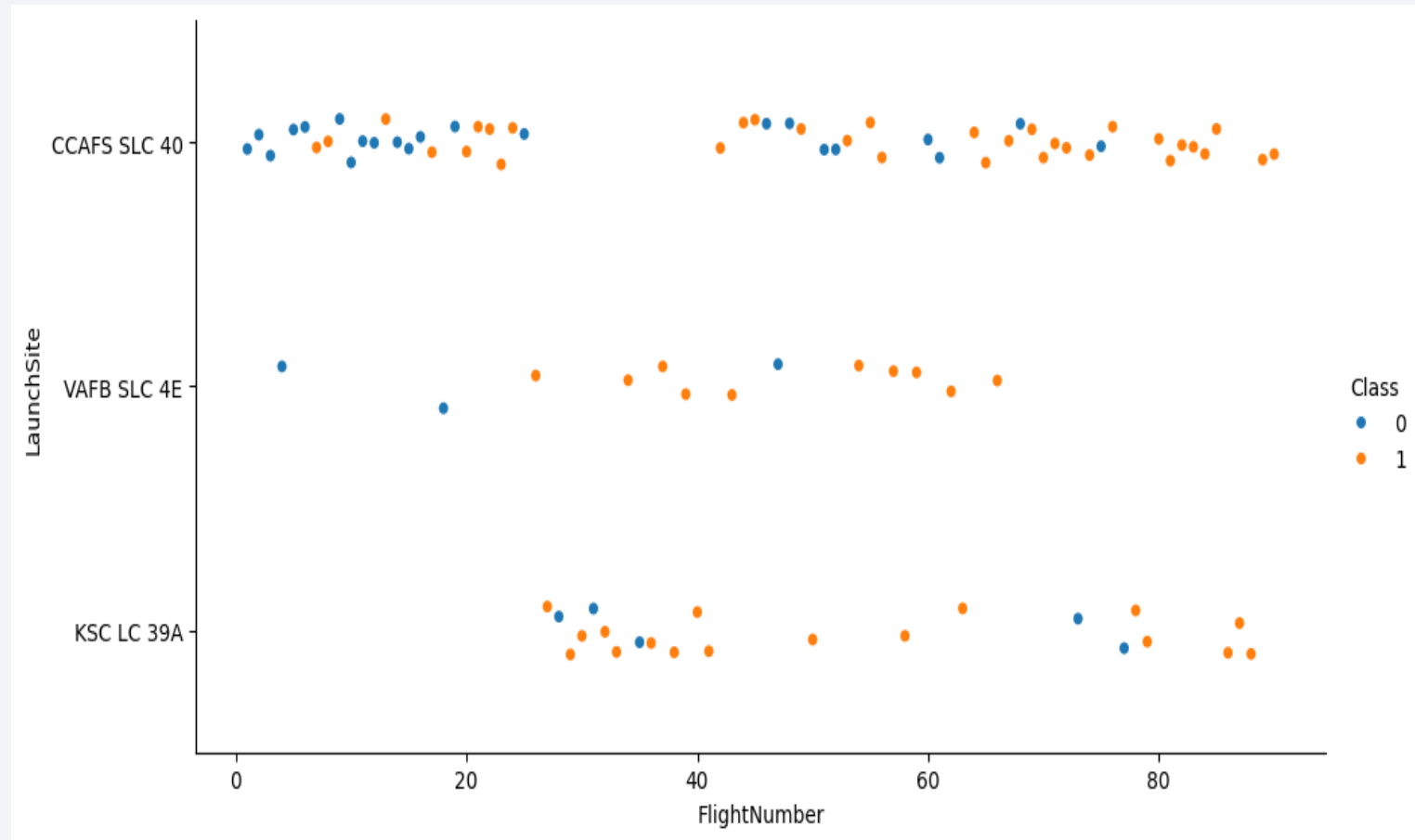


Section 2

Insights drawn from EDA

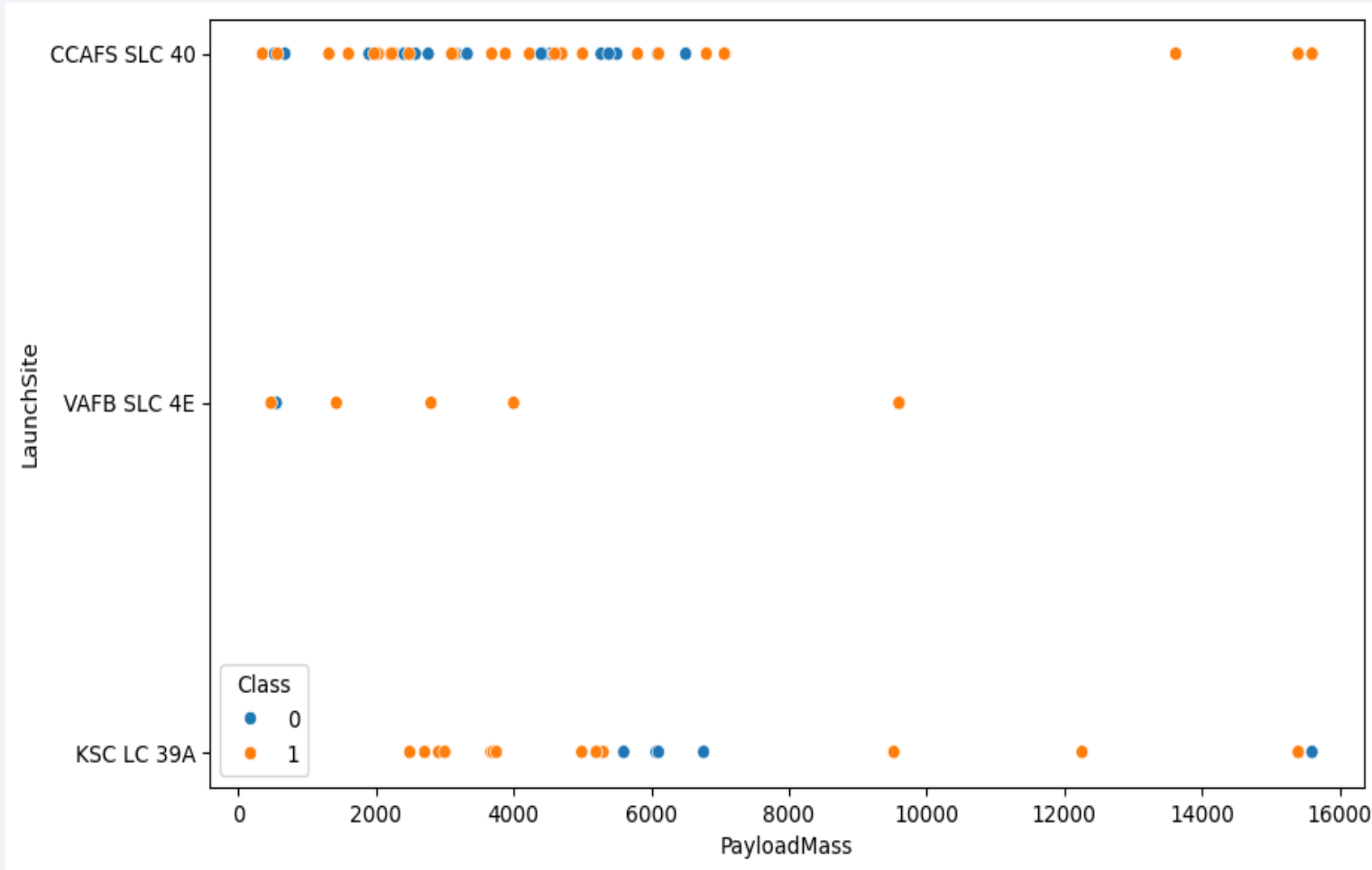
Flight Number vs. Launch Site

- The amount of failed lands was bigger than successful ones in the start, but gradually the amount of successful lands increased and became more with the increasing of flight numbers



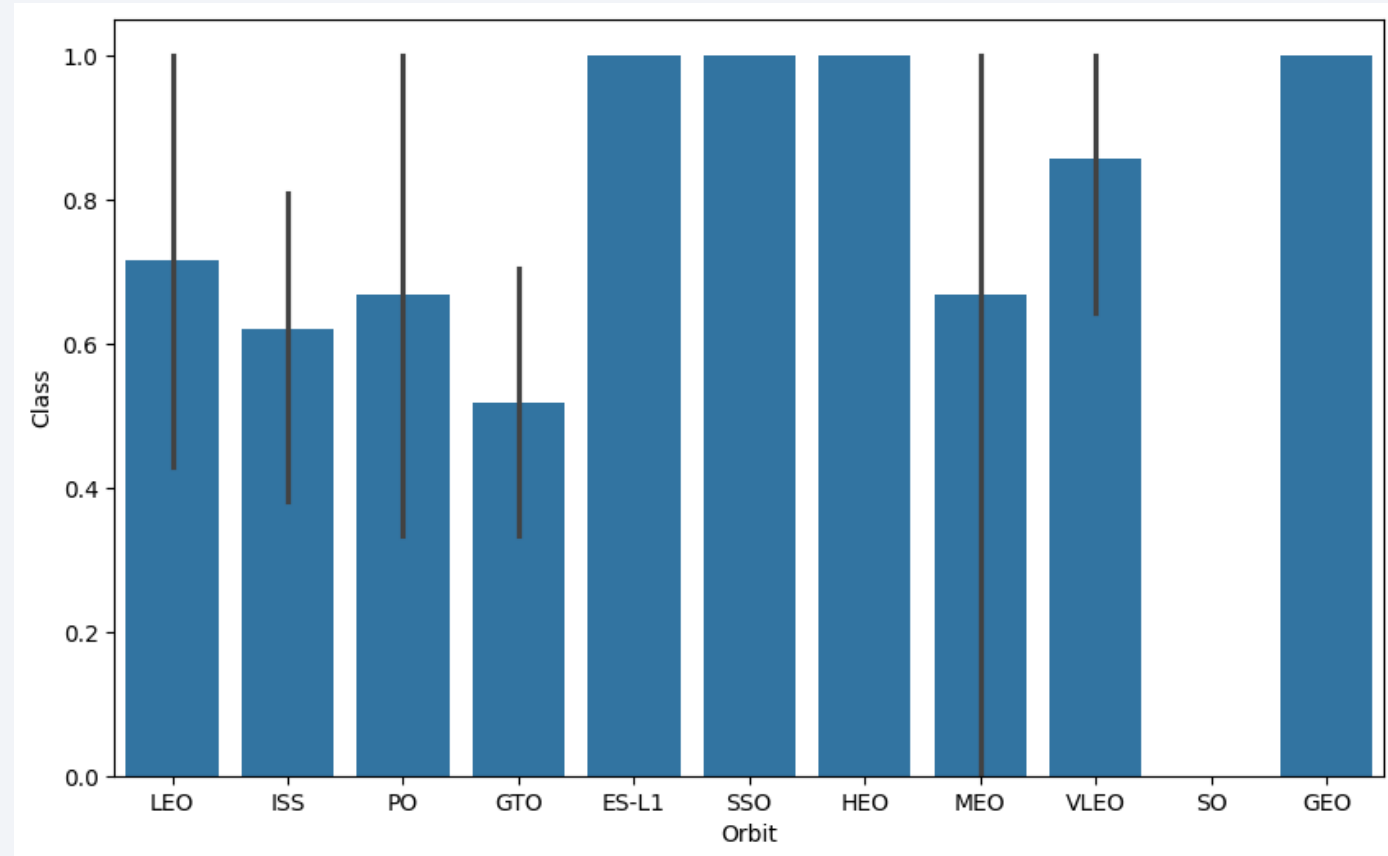
Payload vs. Launch Site

- for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).
- For the CCAFS SLC 40 launchsite it appears that landing outcome and payload mass are not strongly correlated
- For the KSC LC 39A for the range from 5800 to 6700 approximately the landings were not successful but outside of that range almost all landings were successful except for one where PayloadMass almost reached 1600 kg



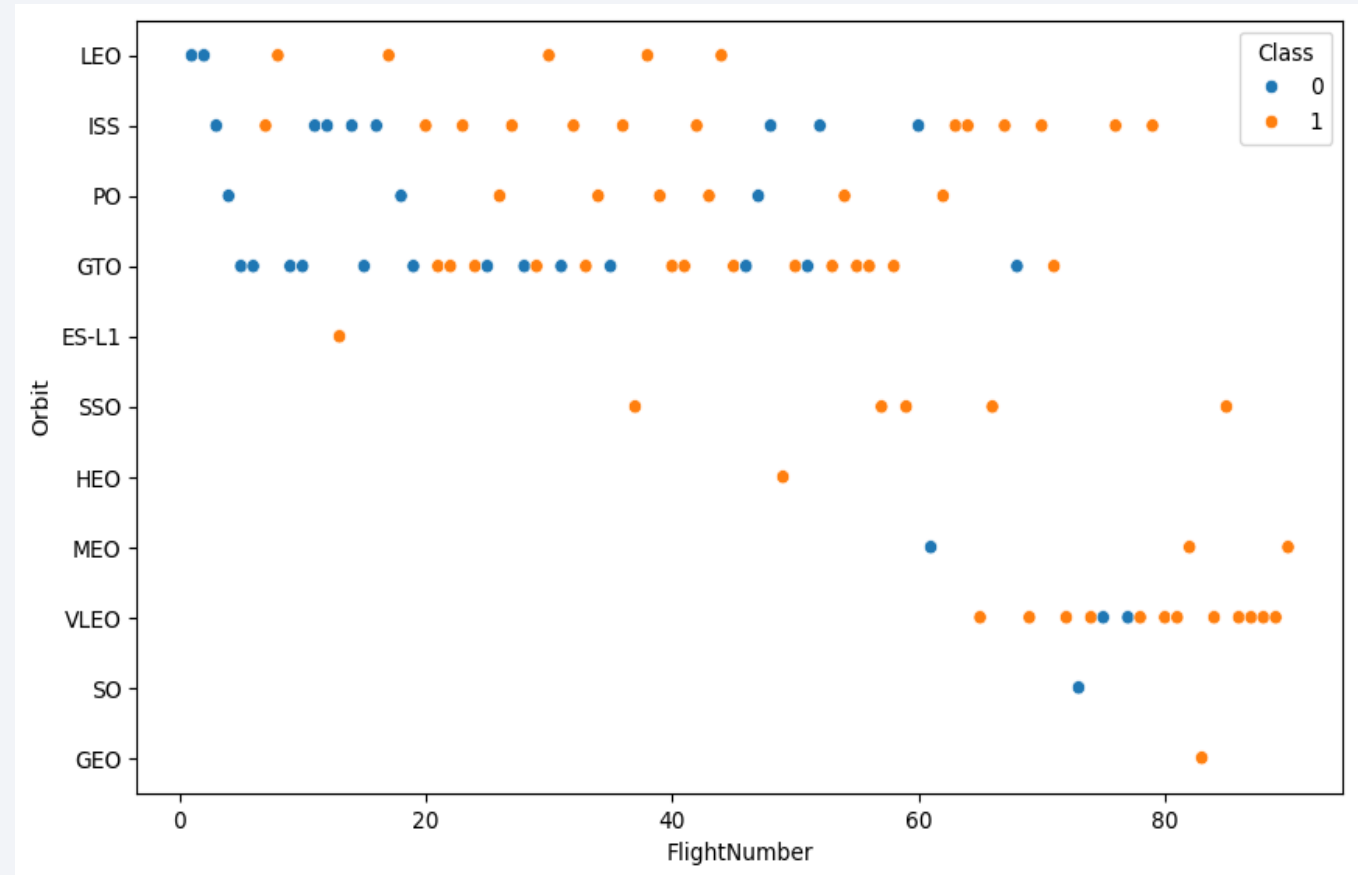
Success Rate vs. Orbit Type

- GEO, ES-L1, and HEO: These orbit types have demonstrated a perfect success rate. However, the limited number of flights for each of these categories makes it challenging to draw definitive conclusions.
- SSO : This orbit type not only shows a perfect success rate but also has a substantial number of flights. This combination suggests a reliably positive trend and provides a more robust basis for evaluating success.
- SO : this orbit didn't show any success yet but it has a low number of flights so the judgement here may not be fair



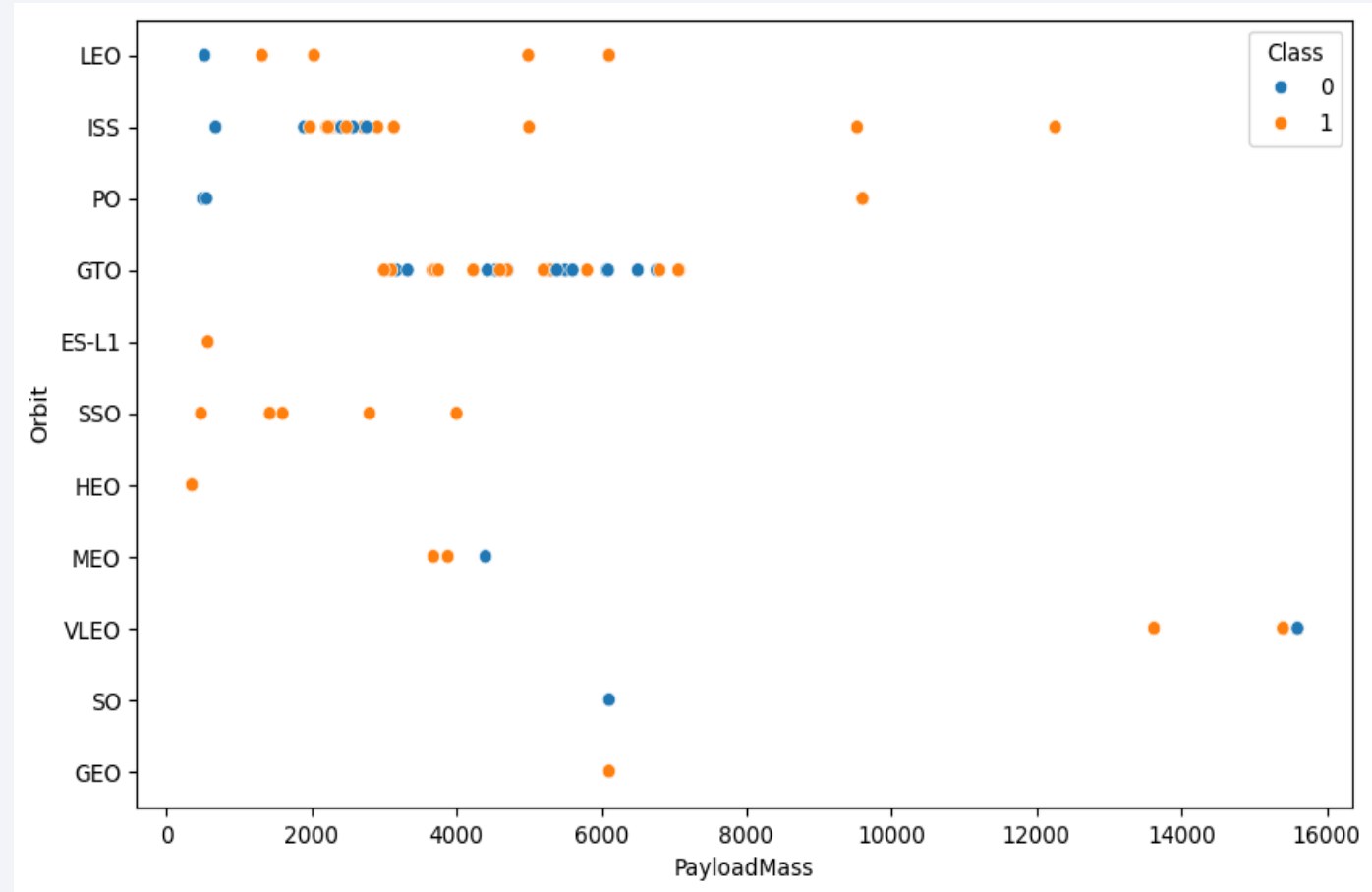
Flight Number vs. Orbit Type

- Most of the orbits had a strong correlation between flight number and success rate, the more the flights made the more success achieved
- e.g., in the LEO orbit the success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number with success rate in GTO orbit.



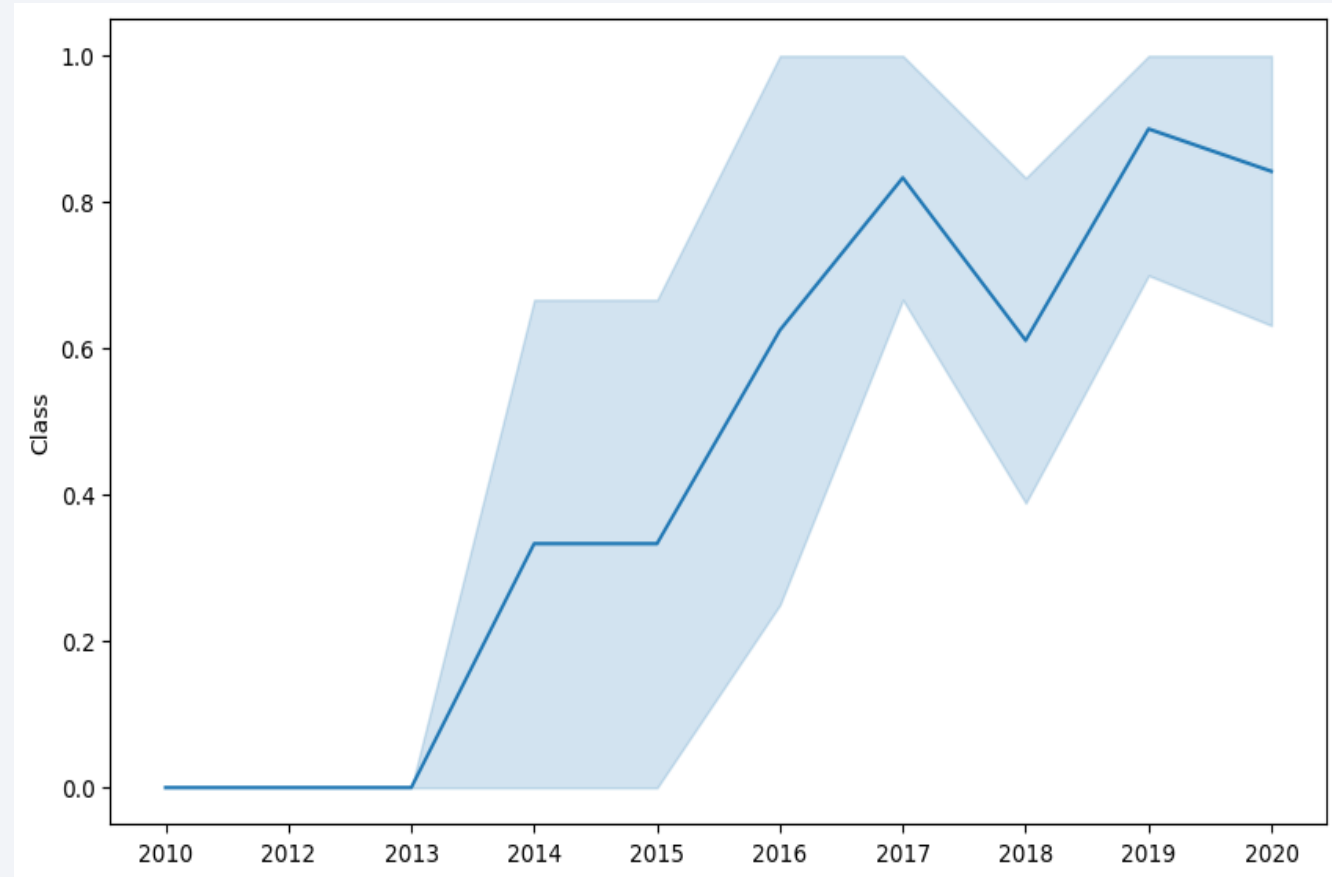
Payload vs. Orbit Type

- payloadMass didn't seem to generally have a clear strong correlation with success nor failure for all orbits
- With heavy payloads the successful landing rate was higher for Polar, LEO and ISS.
- However for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccesful mission) are interchanging



Launch Success Yearly Trend

- The success rate has been increasing since 2013 till 2014 then it became stable till 2015 and started increasing again since approximately 2015 till 2017 and that's when it has dropped till 2018 and returned to increase again till 2019 and went back to dropping in the 2019-2020 period



All Launch Site Names

- There are 4 unique launch sites :
 - CCAFS LC-40
 - VAFB SLC-4E
 - KSC LC-39A
 - CCAFS SLC-40

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT("Launch_Site") FROM SPACEXTBL
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- The query returned 5 records where launch sites begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

✓ 0.0s

Python

* [sqlite:///my_data1.db](#)
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- the total payload carried by boosters from NASA is 45596 kg

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
SELECT SUM("PAYLOAD_MASS__KG_") AS "total payload mass carried by boosters launched by NASA"
FROM SPACEXTBL
WHERE "Customer" = 'NASA (CRS)'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)
Done.

total payload mass carried by boosters launched by NASA
45596

Average Payload Mass by F9 v1.1

- the average payload mass carried by booster version F9 v1.1 is 2928.4 kg

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG("PAYLOAD_MASS__KG_") AS "average_payload_mass_carried_by_booster_version_F9_v1_1"
FROM SPACEXTBL
WHERE "Booster_Version" = 'F9 v1.1'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)
Done.

average_payload_mass_carried_by_booster_version_F9_v1_1
2928.4

First Successful Ground Landing Date

- first successful landing outcome on ground pad happened on 2015-12-22

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%%sql SELECT MIN("Date") as 'the date when the first succesful landing outcome in ground pad was acheived'  
FROM SPACEXTBL  
WHERE "Landing_Outcome" = 'Success (ground pad)'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)
Done.

the date when the first succesful landing outcome in ground pad was acheived

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:
 - F9 FT B1022
 - F9 FT B1026
 - F9 FT B1021.2
 - F9 FT B1031.2

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT "Booster_Version" FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The total number of successful mission outcomes was 100
- The total number of failure mission outcomes was 1

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome",COUNT(*) as Count FROM SPACEXTBL GROUP BY "Mission_Outcome"
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- the names of the booster which have carried the maximum payload mass were :

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
%%sql SELECT "Booster_Version"  
FROM SPACEXTBL  
WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTBL)
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Booster_Version

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015 were:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%%sql SELECT
SUBSTR(Date, 6, 2) AS "Month",
"Landing_Outcome",
"Booster_Version",
"Launch_Site"
FROM
SPACEXTBL
WHERE
"Landing_Outcome" = 'Failure (drone ship)'
AND SUBSTR(Date, 0, 5) = '2015'
```

✓ 0.1s

* [sqlite:///my_data1.db](#)
Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order, the most outcome was no attempt success in general was more than failure and there was a single incident were the outcome was precluded

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%%sql SELECT
    "Landing_Outcome" , COUNT(*) as "Count"
FROM
    SPACEXTBL
WHERE
    Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    "Count" DESC
```

✓ 0.0s

Python

* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Launch Sites Locations

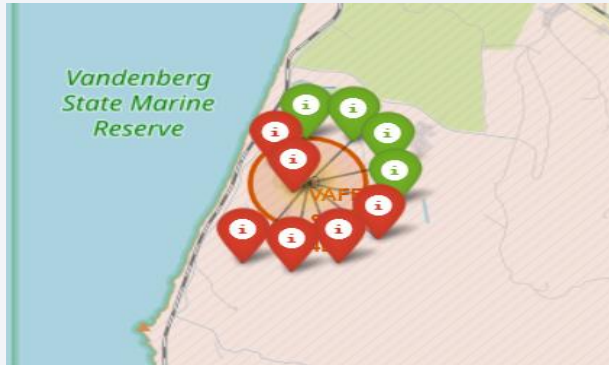
- All sites:
 - VAFB SLC-4E VAFB SLC-4E (Vandenberg Air Force Base, Space Launch Complex 4E)
 - KSC LC-39A (Kennedy Space Center, Launch Complex 39A)
 - CCAFS LC-40 (Cape Canaveral Air Force Station, Launch Complex 40)
 - CCAFS SLC-40 (Cape Canaveral Space Launch Complex 40)

Are the closet they could get to the equator line considering USA border to take advantage of the Earth's rotational speed for increased launch efficiency, while they are also near the coast far from populated areas to minimize the risks and consequences of failures



The success/failed launches for each site

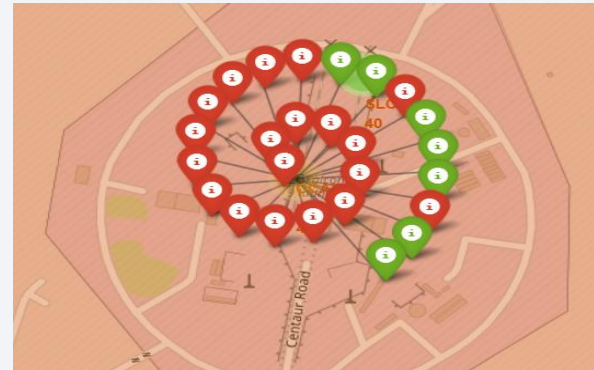
- The markers show the missions outcomes for launches that took place from the launch site
- Green indicates success
- Red indicates failure
- VAFB SLC-4E : had 4 successes, 6 failures
- KSC LC-39A : had 10 successes, 3 failures
- CCAFS LC-40 : had 7 successes, 19 failures
- CCAFS SLC-40 : had 3 success, 4 failures



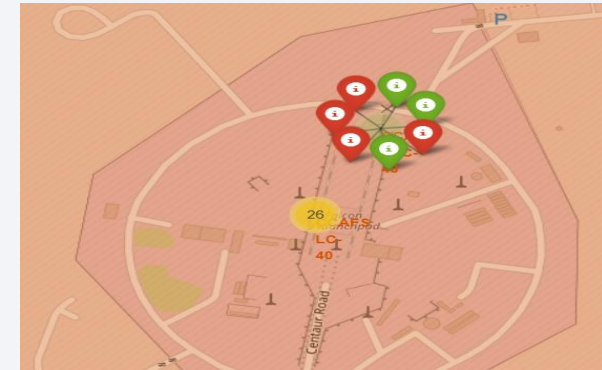
VAFB SLC-4E



KSC LC-39A



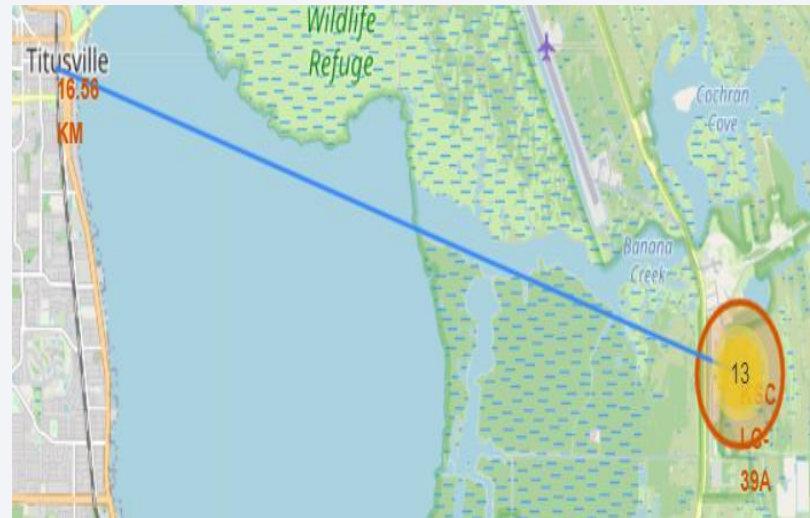
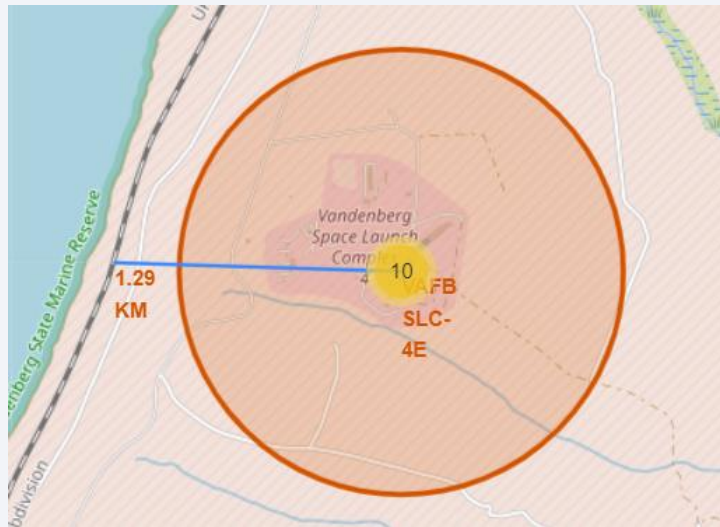
CCAFS LC-40

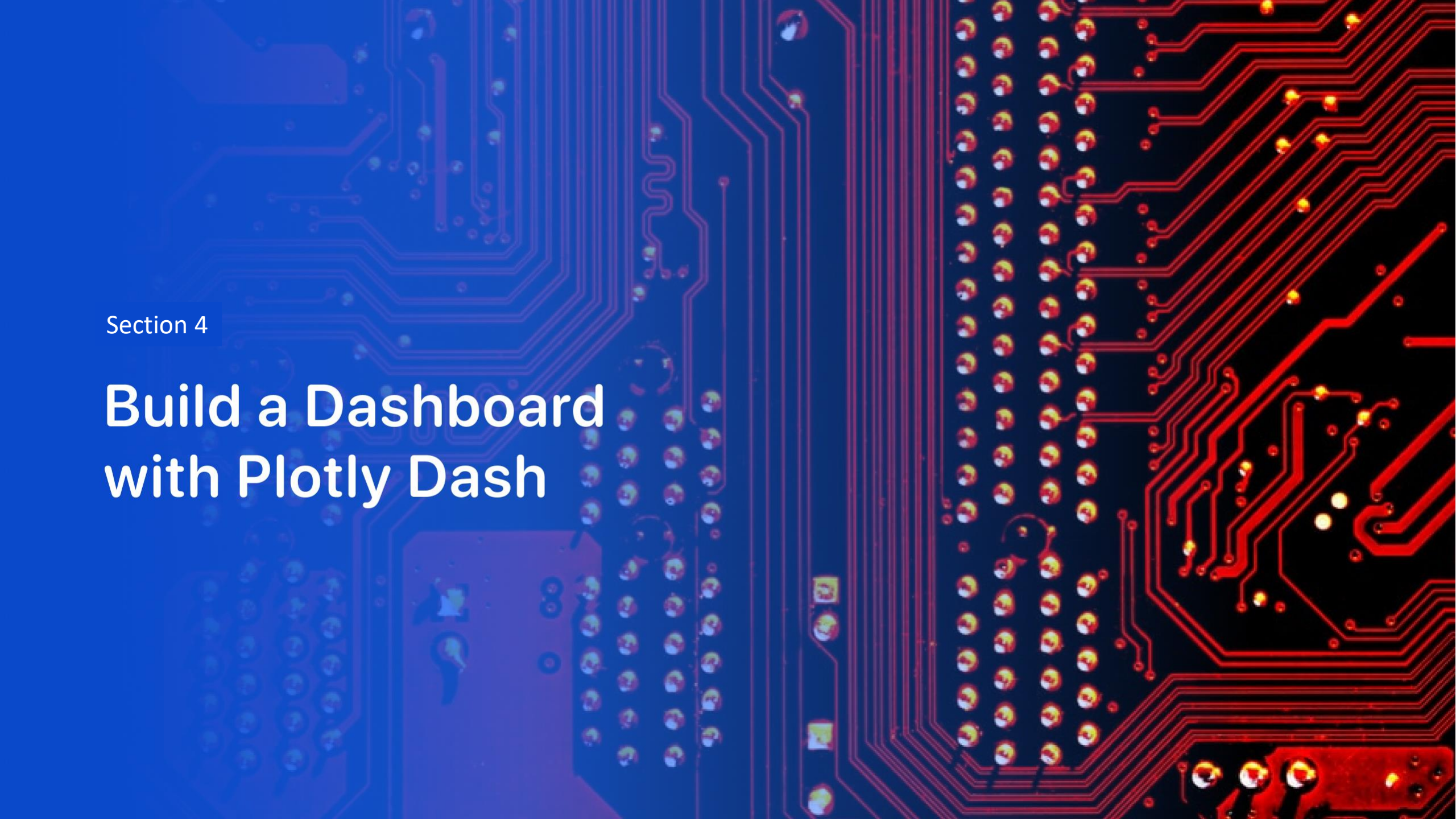


CCAFS SLC-40³⁶

The Distances between a launch site to its proximities

- VAFB SLC-4E is approximately 1.29 KM away from the nearest railway
- KSC LC-39A is approximately 16.56 away from the nearest city
- CCAFS LC-40 is approximately .65 KM away from the nearest railway
- CCAFS SLC-40 is approximately .88 KM away from the nearest coast



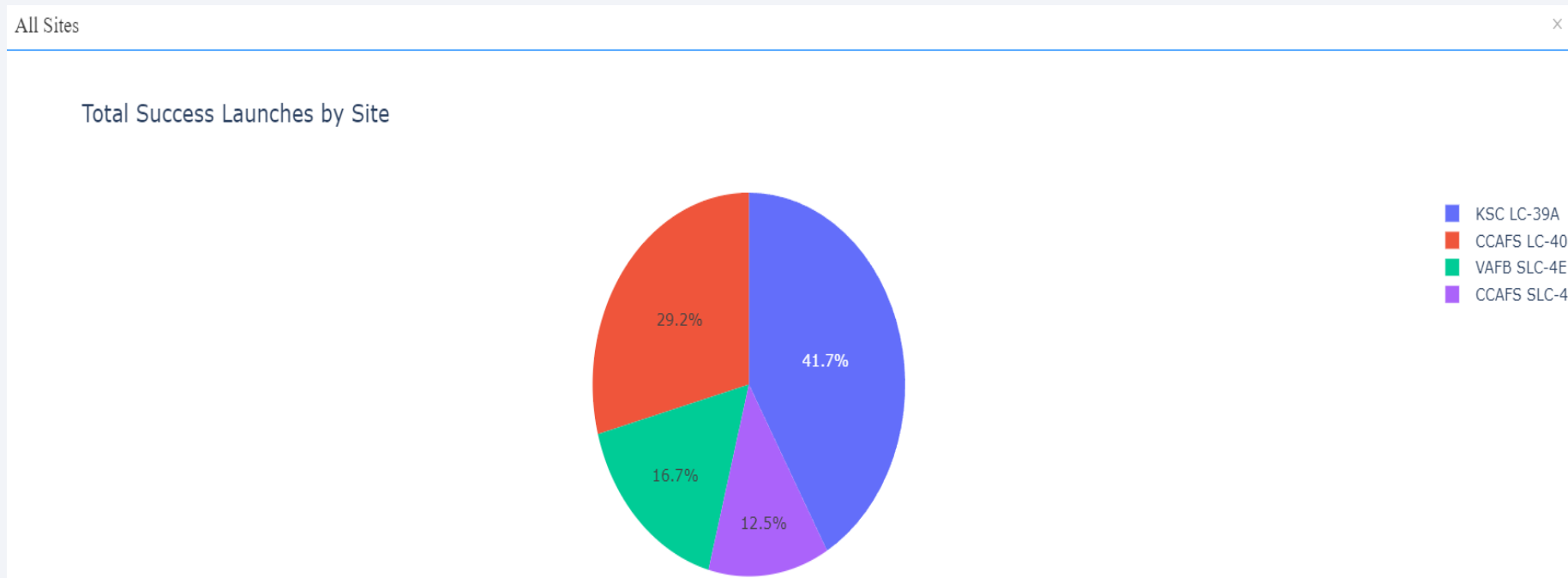


Section 4

Build a Dashboard with Plotly Dash

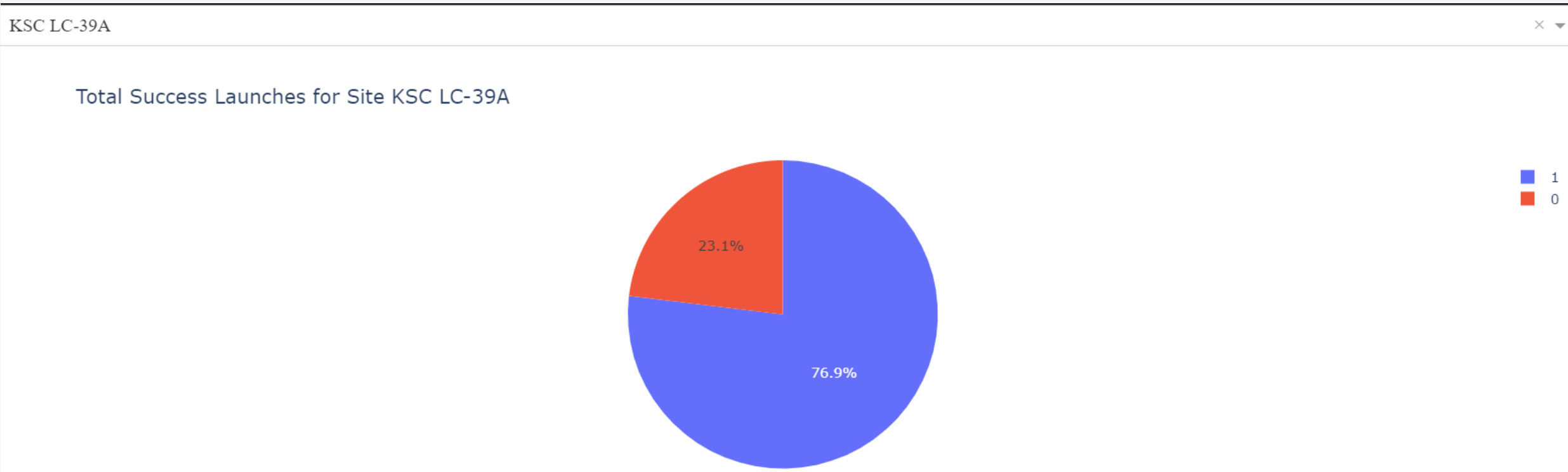
Total Success Launches by Site

- KSC LC-39A site came in 1st with the highest share of success achieved with 41.7% of total successful outcomes
- CCAFS LC-40 site came in 2nd with 29.2% of total successful outcomes
- VAFB SLC-4E site came in 3rd with 16.7% of total successful outcomes
- CCAFS SLC-40 site came in 4th with 12.5% of total successful outcomes



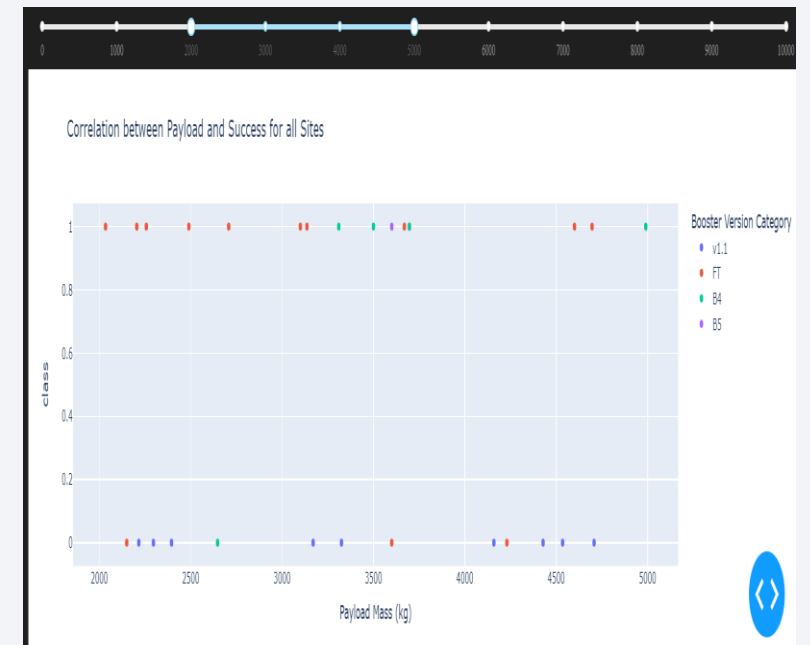
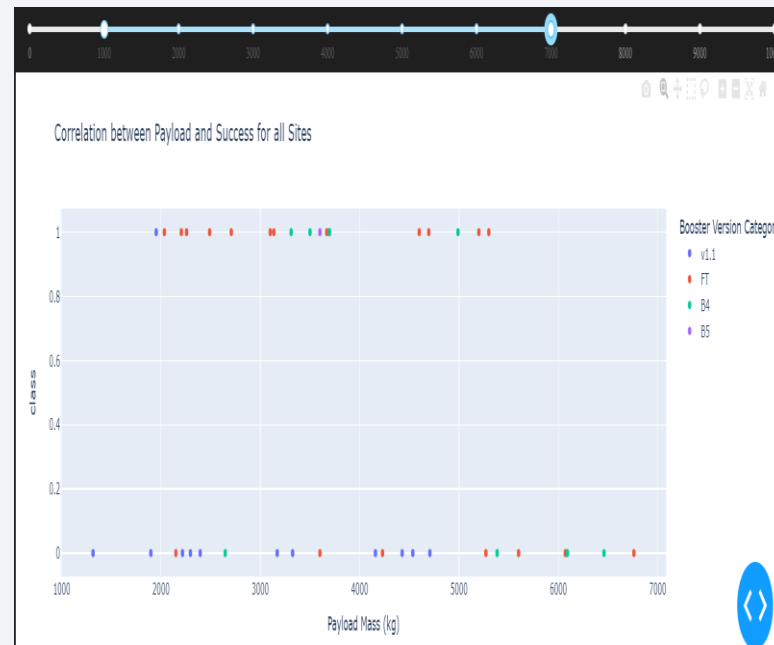
The Site With The Highest Success Rate

- KSC LC-39A Site was the site with the highest success rate
- Launches from KSC LC-39A Site are successful 76.9% of the time and failed 23.1% of the time



Correlation between payload and success for all sites

- The payload range with the largest success rate was 2000 to 5000 kg
- The booster version with the largest success rate was FT

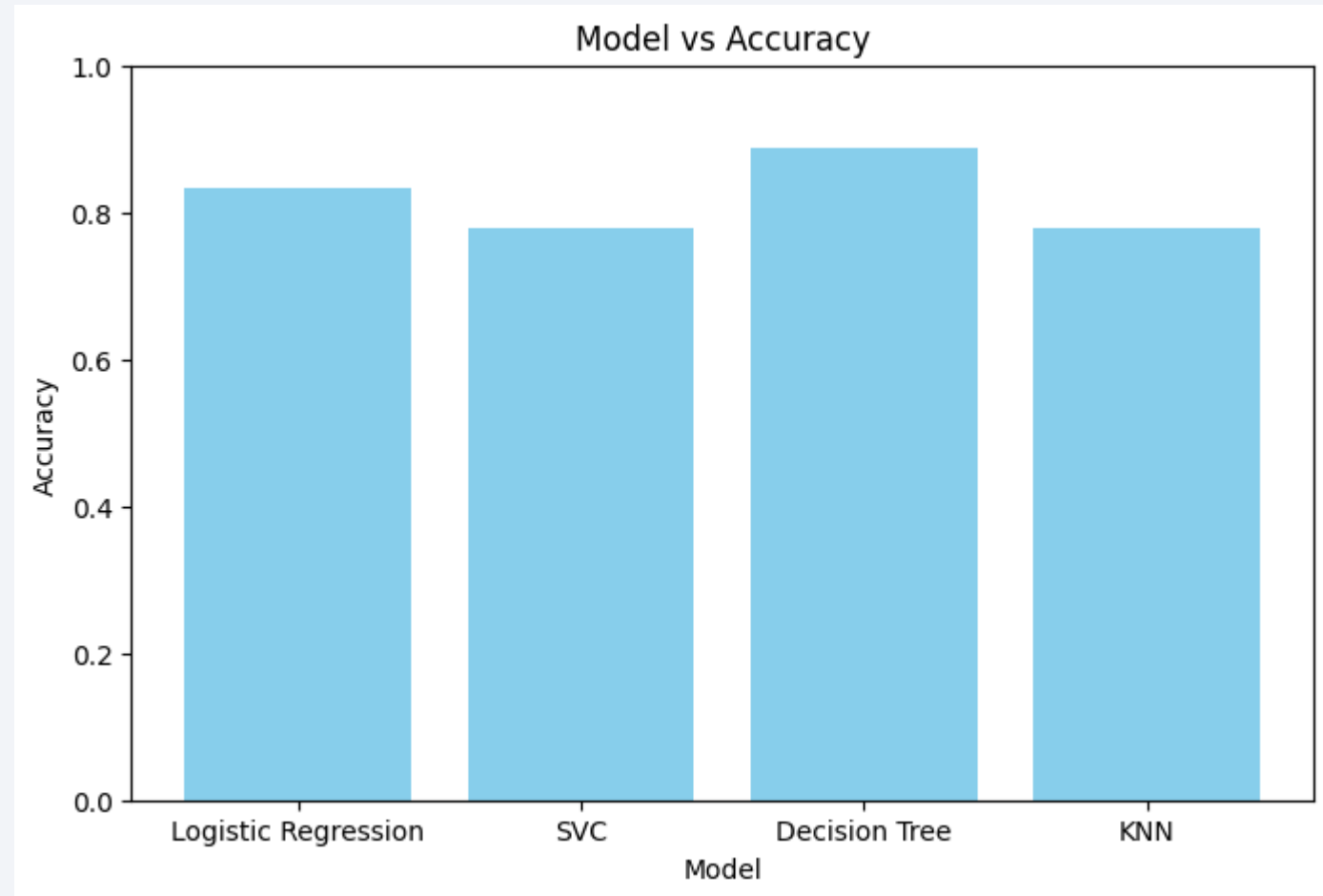


Section 5

Predictive Analysis (Classification)

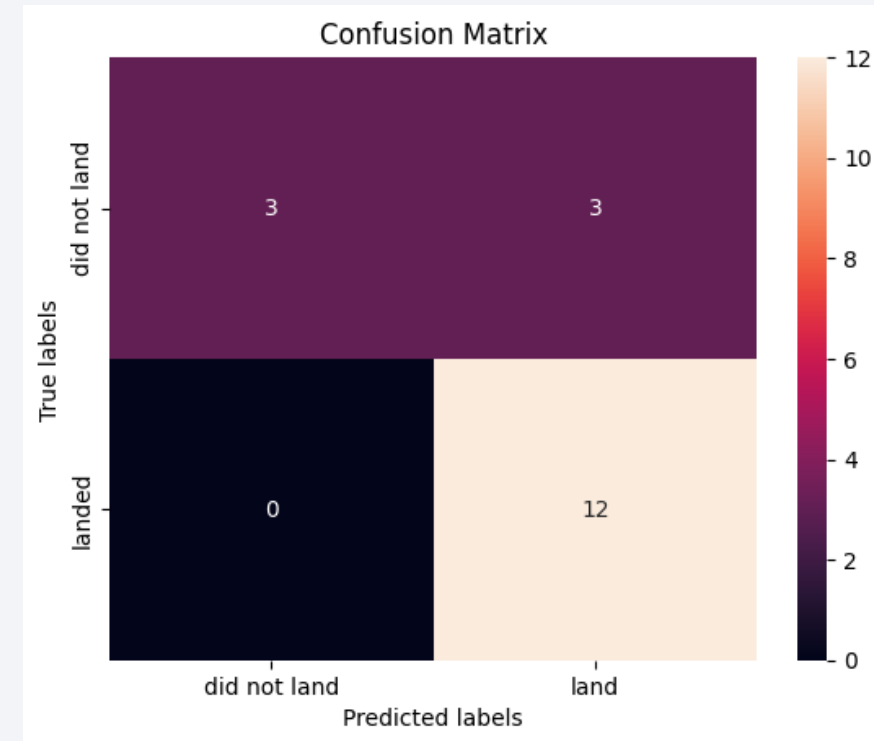
Classification Accuracy

- The model with highest classification accuracy was Decision Tree



Confusion Matrix

- The best performing model was logistic regression since it had the best F2 Score, lowest Log Loss, and competitive Jaccard and Accuracy.
- Explanation of the confusion matrix:
 - The model predicted “land” correctly 12 times since the actual label was “landed”
 - The model predicted “land” falsely 3 times since the actual label was “did not land”
 - The model predicted “did not land” correctly 3 times since the actual label was “did not land”
 - The model never predicted “did not land” when the actual label was “landed”



Conclusions

- Success rate increased over the years and with the increase of number of flights which indicates a good progress
- Orbits such as ES-L1, SSO, HEO and GEO had high success rates
- KSC LC-39A site was the launch site with the highest success rate
- FT booster version category had the highest success rate amongst all other booster version categories
- ML models such as logistic regression and decision tree can be used to predict the landing outcome

Appendix

- Python code snippets, SQL queries, charts, Notebook outputs have been included through the representation
- The link to the main project github repository : [OmarHani4306/Applied-Data-Science-Capstone](https://github.com/OmarHani4306/Applied-Data-Science-Capstone)
- The links for data sets created during the project :
 - [Applied-Data-Science-Capstone/dataset_part_1.csv at main · OmarHani4306/Applied-Data-Science-Capstone](#)
 - [Applied-Data-Science-Capstone/dataset_part_2.csv at main · OmarHani4306/Applied-Data-Science-Capstone](#)
 - [Applied-Data-Science-Capstone/dataset_part_3.csv at main · OmarHani4306/Applied-Data-Science-Capstone](#)
 - [Applied-Data-Science-Capstone/spacex_launch_geo.csv at main · OmarHani4306/Applied-Data-Science-Capstone](#)
 - [Applied-Data-Science-Capstone/spacex_web_scraped.csv at main · OmarHani4306/Applied-Data-Science-Capstone](#)

Thank you!

